

Innovation Summary

Project Title : Brain-Controlled Verbal Communication Enabled Wheelchair

Overview

The Brain-Controlled Verbal Communication Enabled Wheelchair is an affordable assistive mobility solution designed for individuals suffering from severe physical disabilities, paralysis, ALS, spinal cord injuries, and speech impairments. The system enables users to navigate the wheelchair using EEG-based eye blink commands while simultaneously providing a verbal communication interface for expressing basic needs.

Unlike conventional smart wheelchairs that are expensive and dependent on cloud-based processing, our solution performs real-time processing on edge devices, making it suitable for both indoor and outdoor environments with limited or no internet connectivity.

Key Innovations:

- Real-time EEG-based wheelchair navigation using eye blink detection (EEG Signals).
- Edge AI processing for low-latency operation.
- Integrated obstacle detection using ultrasonic sensors.
- Affordable implementation using commercially available hardware.
- Modular architecture for future feature expansion.
- Assistive verbal communication for speech-impaired users.
- Low power consumption and portable design.

Impact:

The proposed system aims to improve the independence, safety, and communication abilities of physically challenged individuals while maintaining affordability compared to existing commercial Brain-Computer Interface (BCI) wheelchairs.